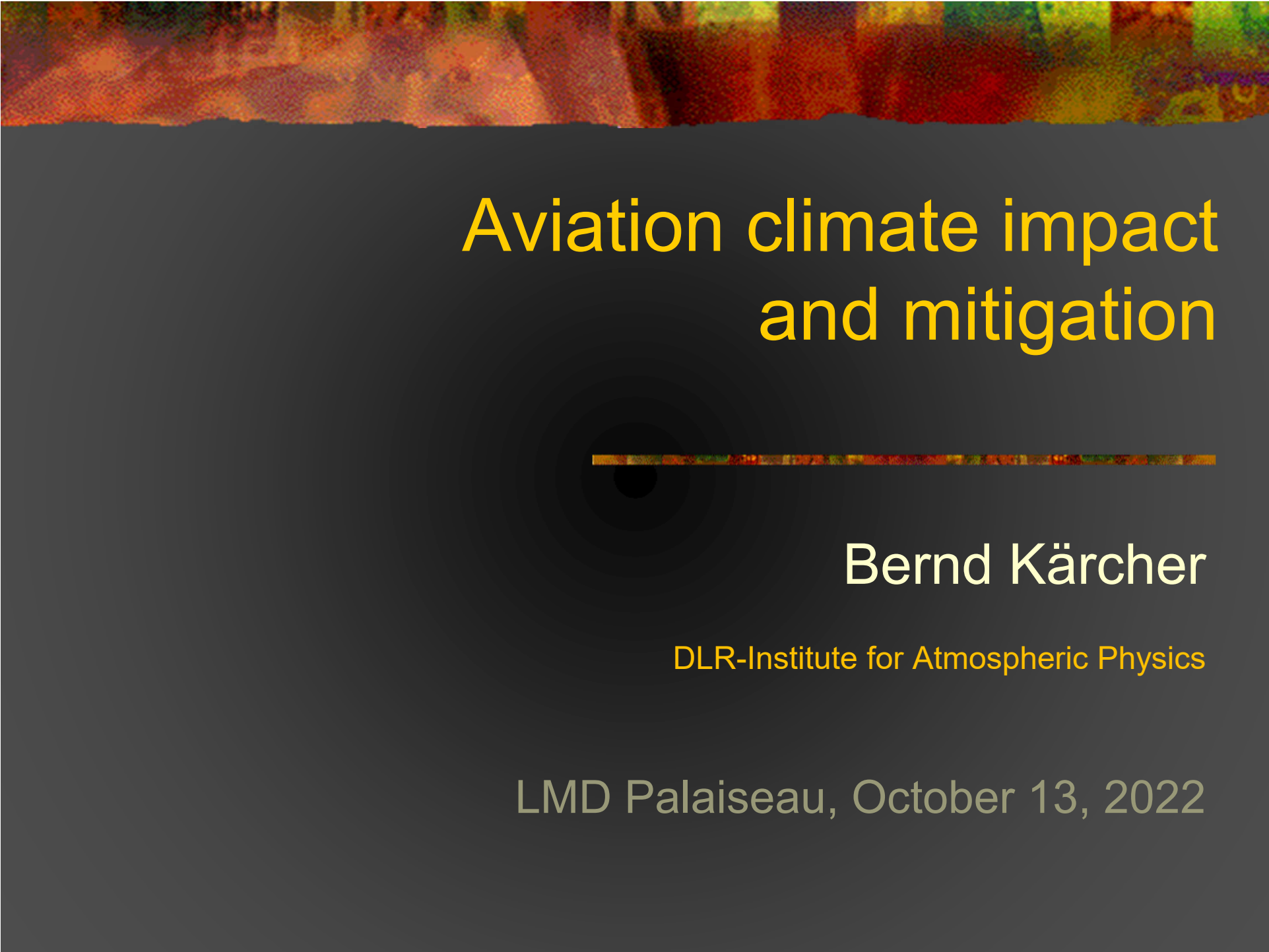




Aviation climate impact and mitigation

Bernd Kärcher

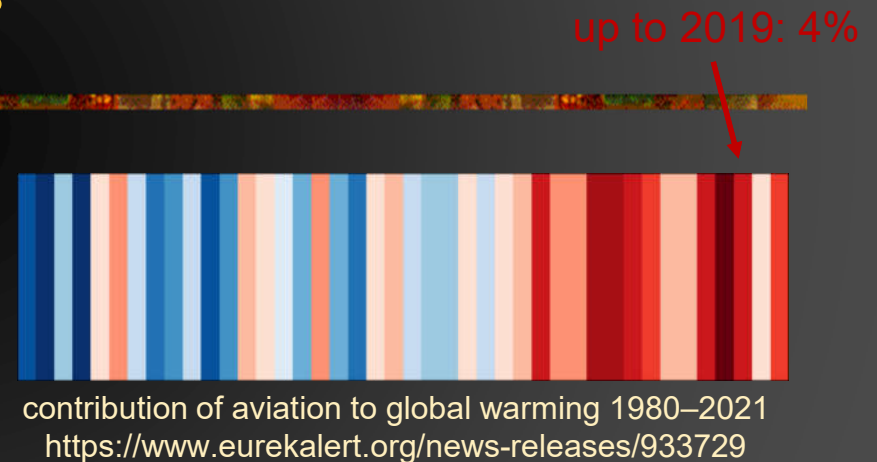
DLR-Institute for Atmospheric Physics

LMD Palaiseau, October 13, 2022

Global Warming (IPCC SR 15, 2018)

- Earth has warmed globally (since 1850–1900) already by $\approx 1.1^\circ\text{C}$; limiting global warming to 1.5°C by 2100 requires a **dramatic pace for decarbonization** across all sectors

- 1.5°C temperature guard rail could be exceeded already in 2030
- every tenth of a degree less warming matters



- aviation is the **most carbon-intensive** way to travel (emission per person):
 - single round trip Munich — Washington D.C. equivalent to **1.3 t** emissions of CO_2
 - average annual CO_2 emissions in Germany (in 2019) amount to **8.5 t** per person

Wright brothers' fixed-wing aircraft —
first controlled powered flight ever (Kitty Hawk, NC)

12 hp (9 kW) • 12 ft/s (13 km/h)
Dec 17, 1903



Aviation CO₂ and contrail clouds

already CO₂ emissions but no contrails

1903



today

accumulated CO₂ emissions and contrail clouds



Aviation-induced clouds (AIC)



young line-shaped **contrails** and the older irregularly-shaped **contrail cirrus** clouds that arise from them (WMO: *Ci homogenitus*)

Short- vs long-lived forcing agents



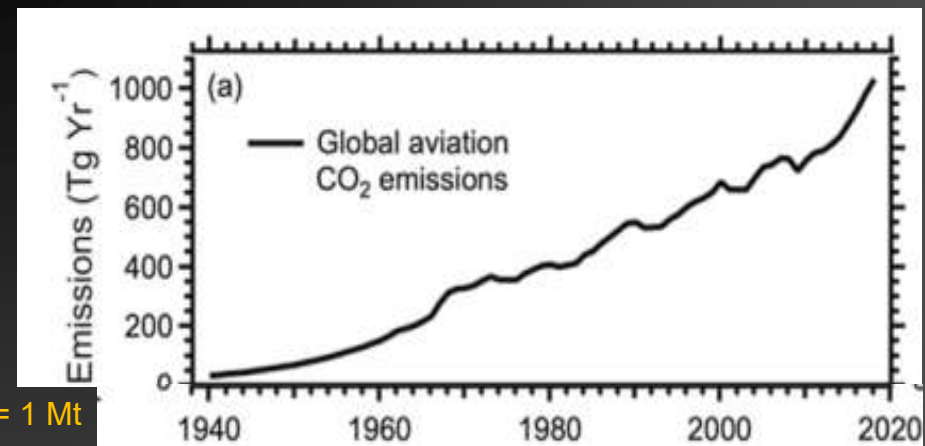
Cloud effects on the atmospheric moisture and temperature budgets **are transient.**

William Turner (1775–1851)
~20,000 paintings in 76 years

Temporal history of CO₂ emissions must be known to estimate their effect on climate:

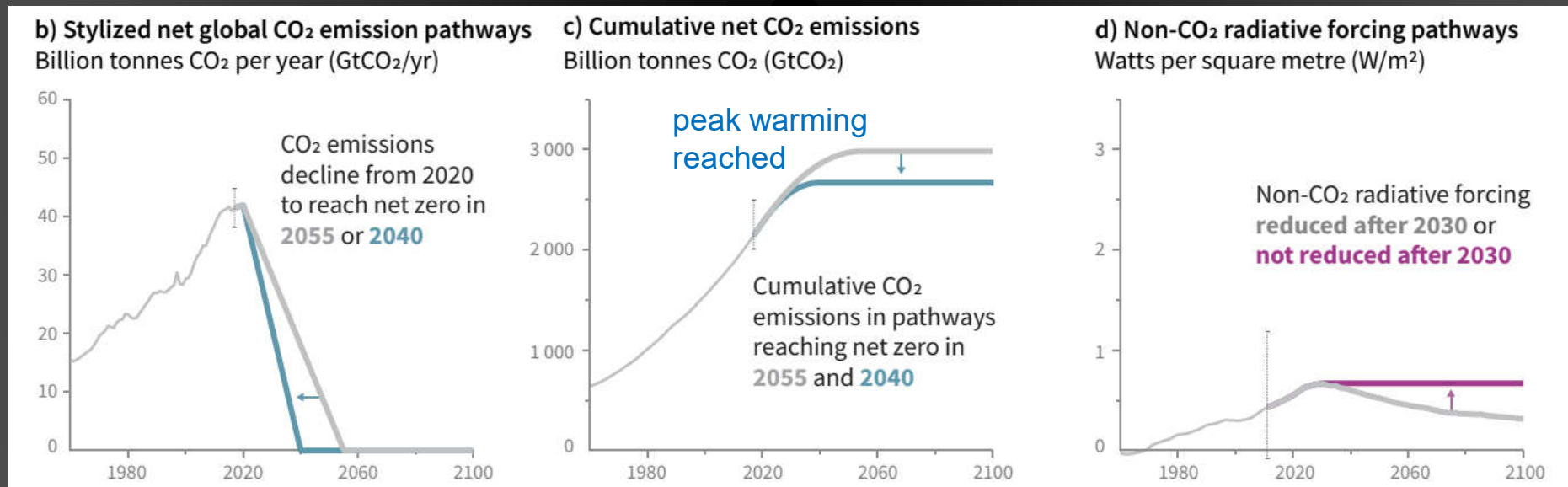
- CO₂ is long-lived and mixes rapidly within and across hemispheres
- 20% remain in atmosphere for millennia (slow carbon sinks in oceans)

1 Tg = 1 Mt



The importance of 'net zero'

- Net zero is achieved when anthropogenic CO₂ **emissions are balanced by anthropogenic CO₂ removals** globally over a specified period
- Reaching and sustaining net zero *and* declining net non-CO₂ RF* halts anthropogenic global warming on multi-decadal time scales

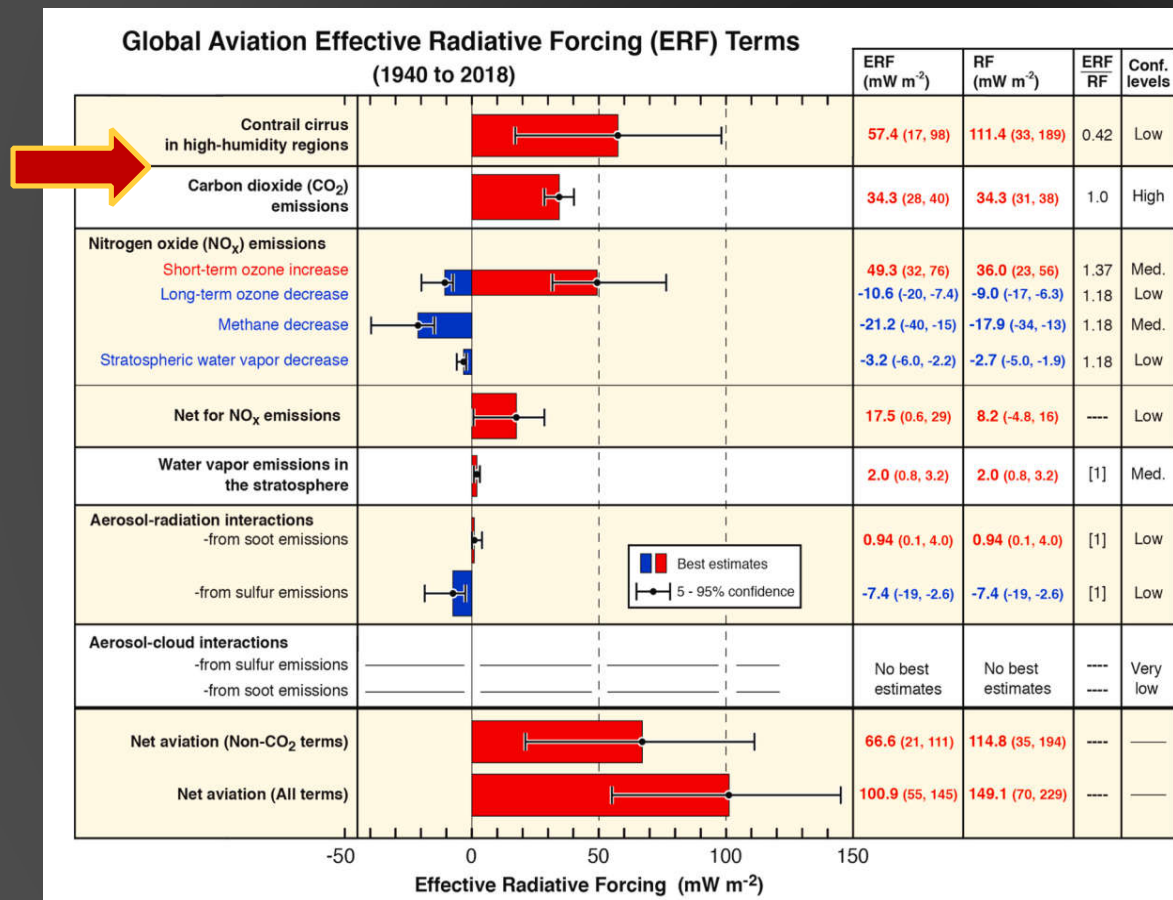


*RF (simplified): radiative flux imbalance at TOA



Aviation's contribution to global warming

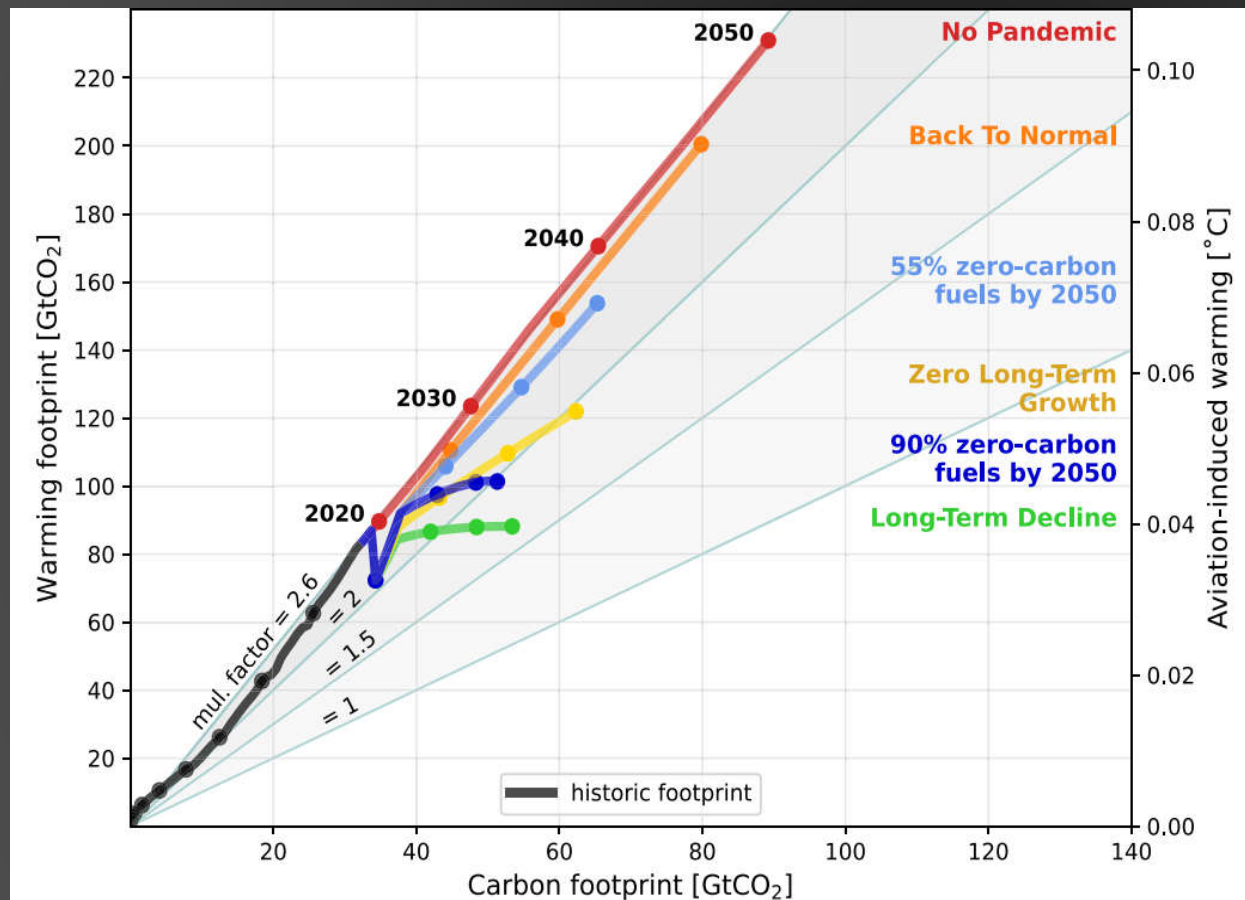
Latest aviation assessment (2018)



- major aviation forcing agents: CO₂ & AIC
- to date, almost **2/3** of total ERF* due to non-CO₂ effects
- low efficacy of AIC to force temperature response (as ratio ERF/RF < 1)
- non-CO₂ effects contribute **8** times more to overall uncertainty than CO₂

*ERF: better metric than RF to estimate impact on surface T changes

Warming footprint of aviation



Back to normal after COVID19:
aviation caused by 2050 $\approx 0.09^\circ\text{C}$
= up to 1/6 of remaining global
warming (COP21 target)

Zero growth leads to a warming
of 0.05°C by 2050

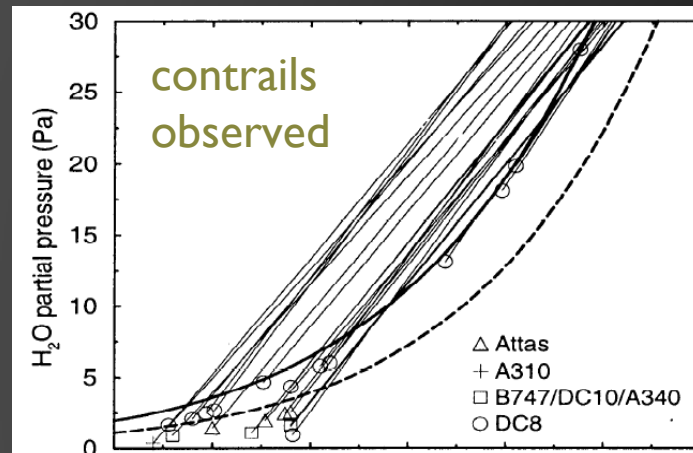
Science behind contrail formation



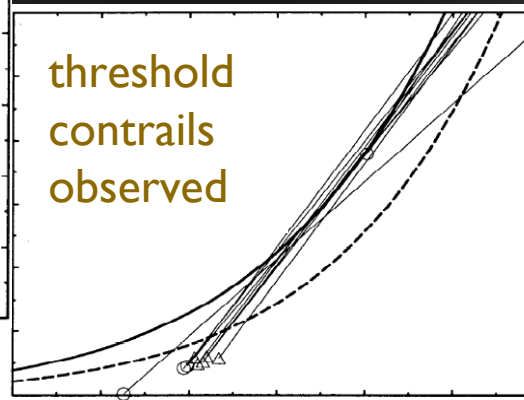
Contrail occurrence

Graphs compare thermodynamic mixing lines and saturation vapor pressures in expanding exhaust plumes:

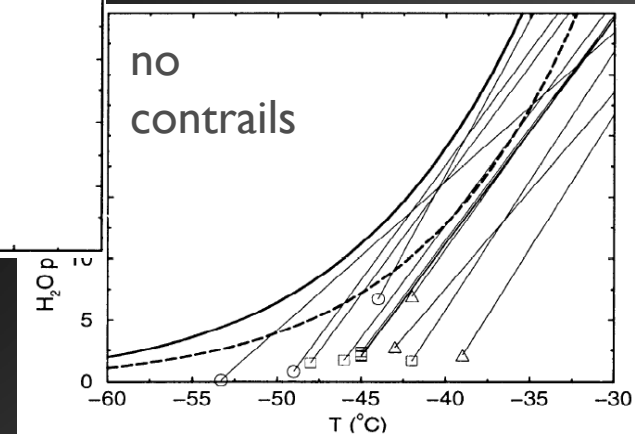
Data show that **contrails form like liquid water clouds**



A high number of small, ice-phase particles form, ensuring early contrail visibility and allowing contrails to persist in ice-supersaturated air



Formation threshold temperatures are solely determined by the mixing line slope; contrails have fewer, larger ice crystals



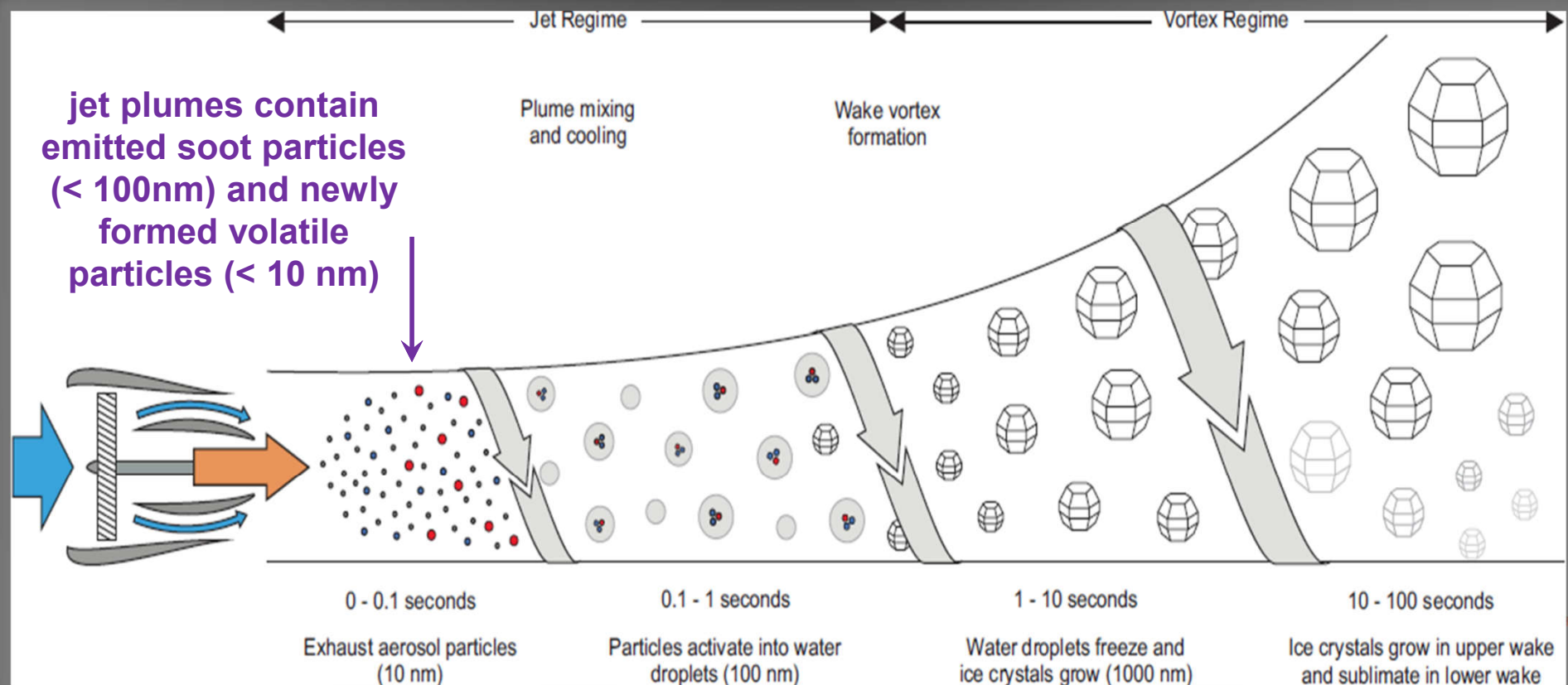
Below those threshold temperatures, no contrail ice crystals form

- liquid water saturation
- - - ice saturation

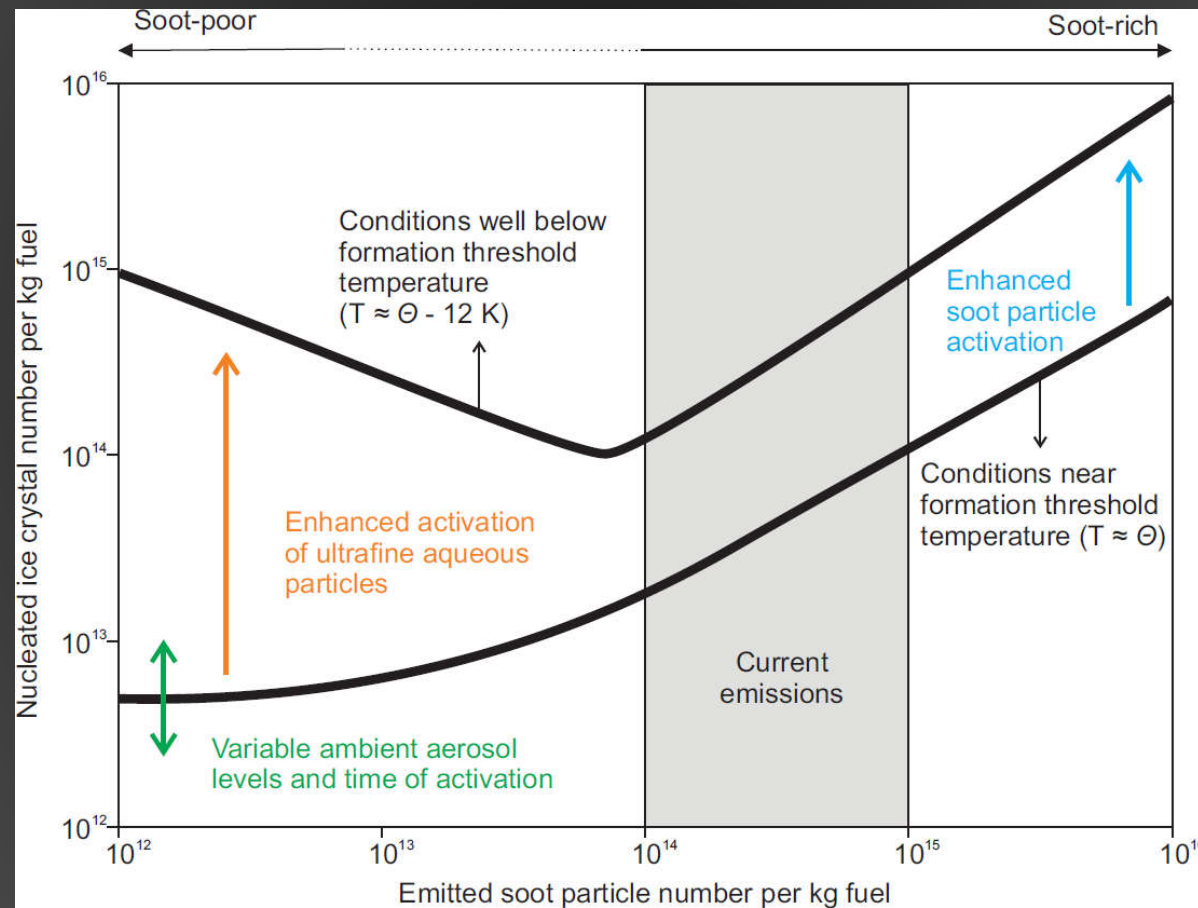
Stages of AIC evolution

aircraft-dominated contrail formation stage has a large potential to support the mitigation of aviation's climate impact

Formation stage processes

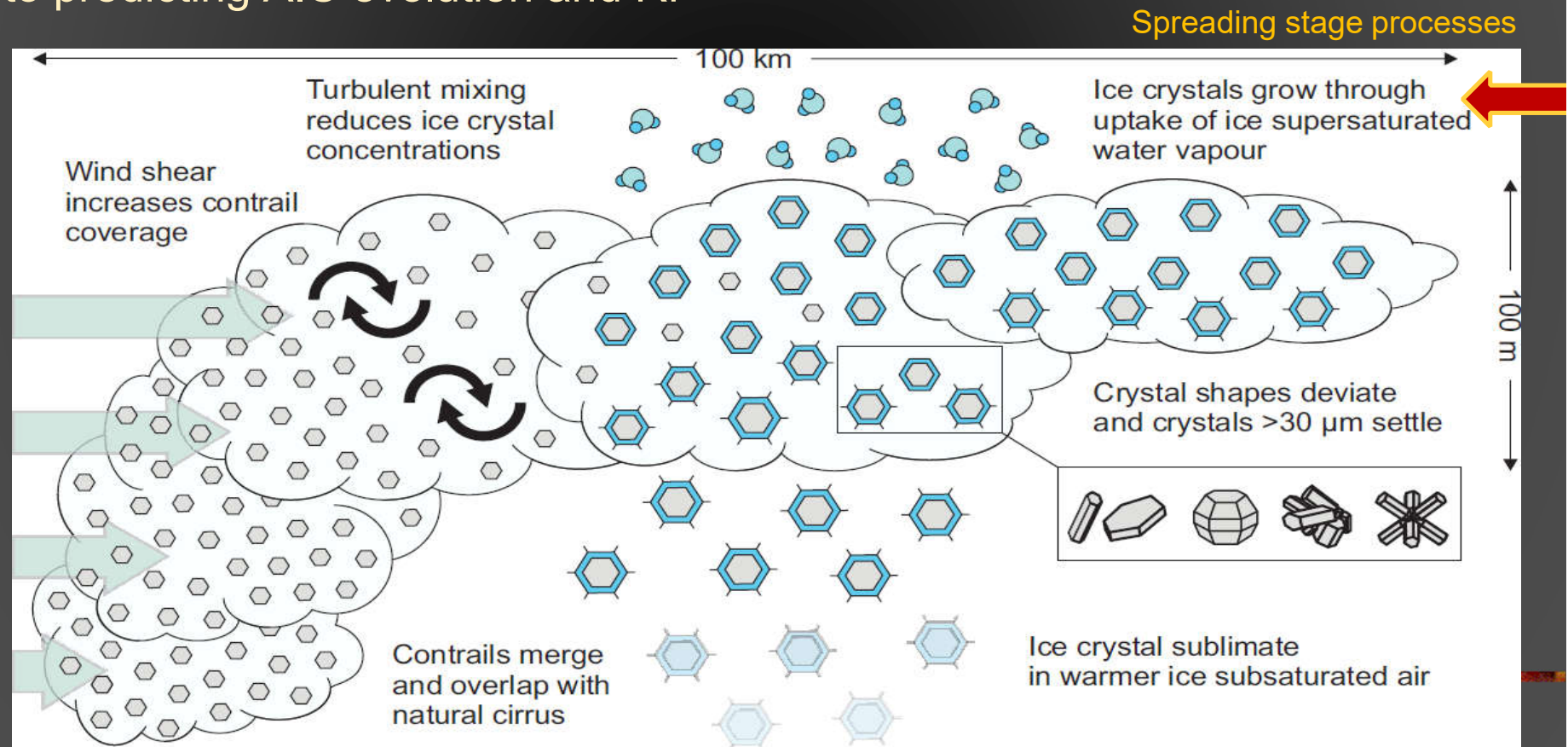


Contrail ice crystal nucleation

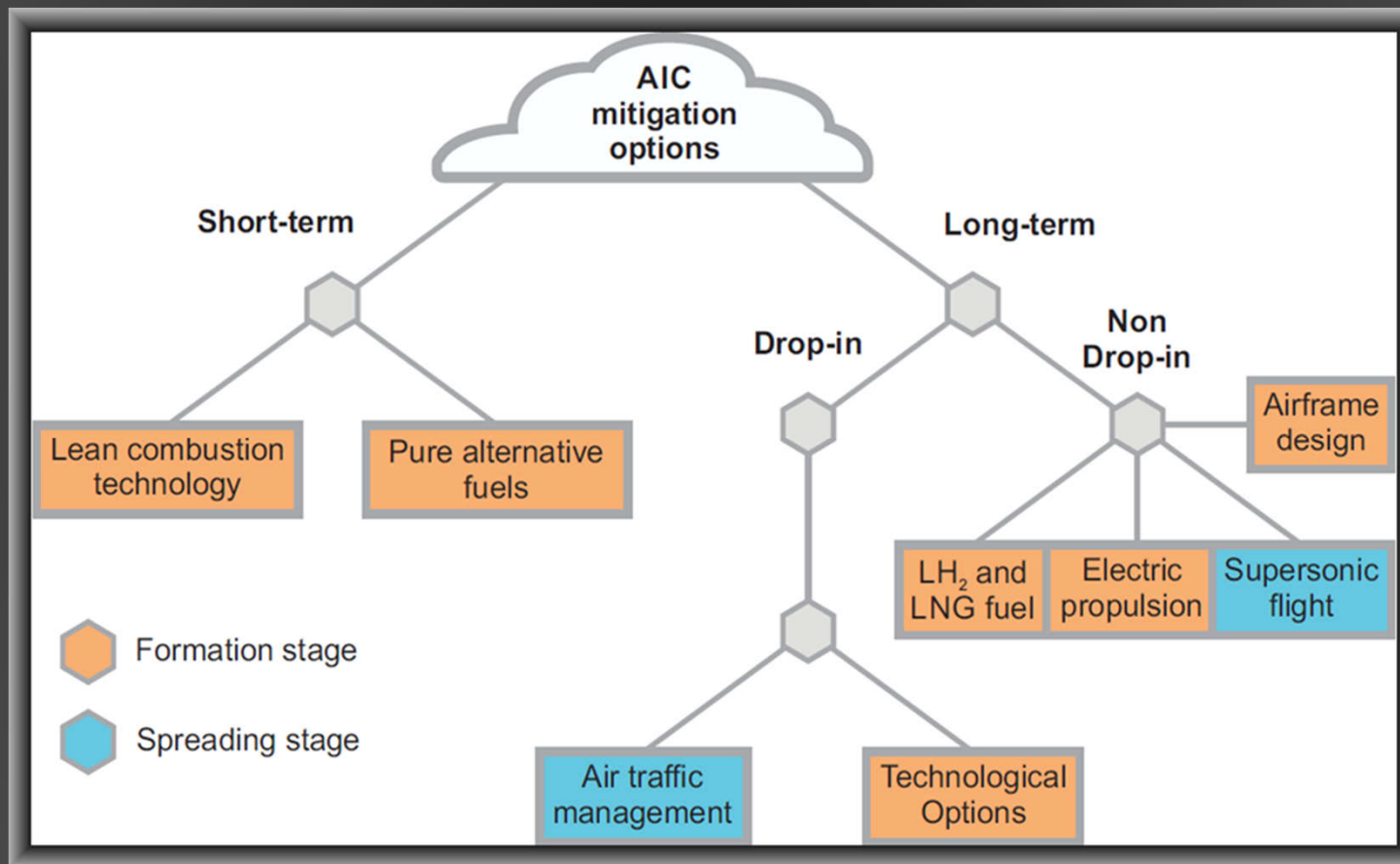


Stages of AIC evolution

understanding atmospheric spreading stage processes is essential to predicting AIC evolution and RF



Mitigating the climate impact



Towards mitigation

- all mitigation options must be 'green' (zero lifecycle CO₂ emissions)
- mitigation of non-CO₂ effects must not enhance CO₂ emissions
- uncertainties in non-CO₂ effects must be taken into account

■ Short-term Options (up to 2035)

- sustainable aviation fuels (possibly combined with lean combustion)
- navigational contrail avoidance

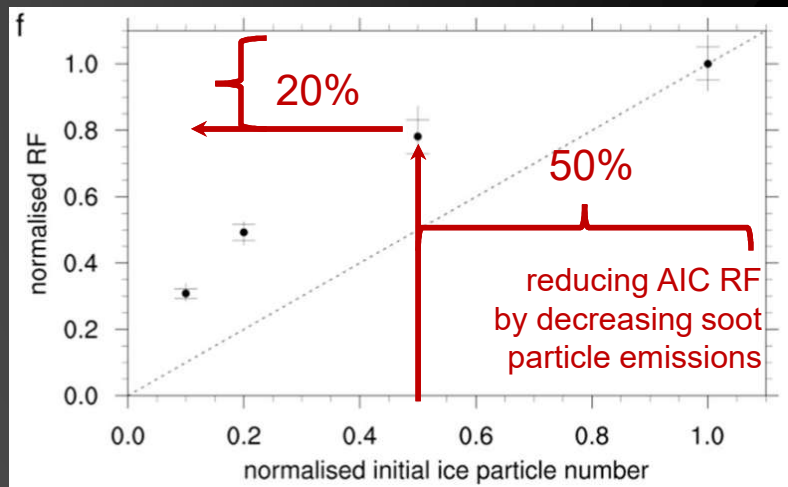
■ Long-term options (beyond 2050)

- radically new aircraft designs
 - liquid hydrogen fuel
 - all-electric aircraft / fuel cells
-

Sustainable aviation fuels

biofuels & power-to-liquid

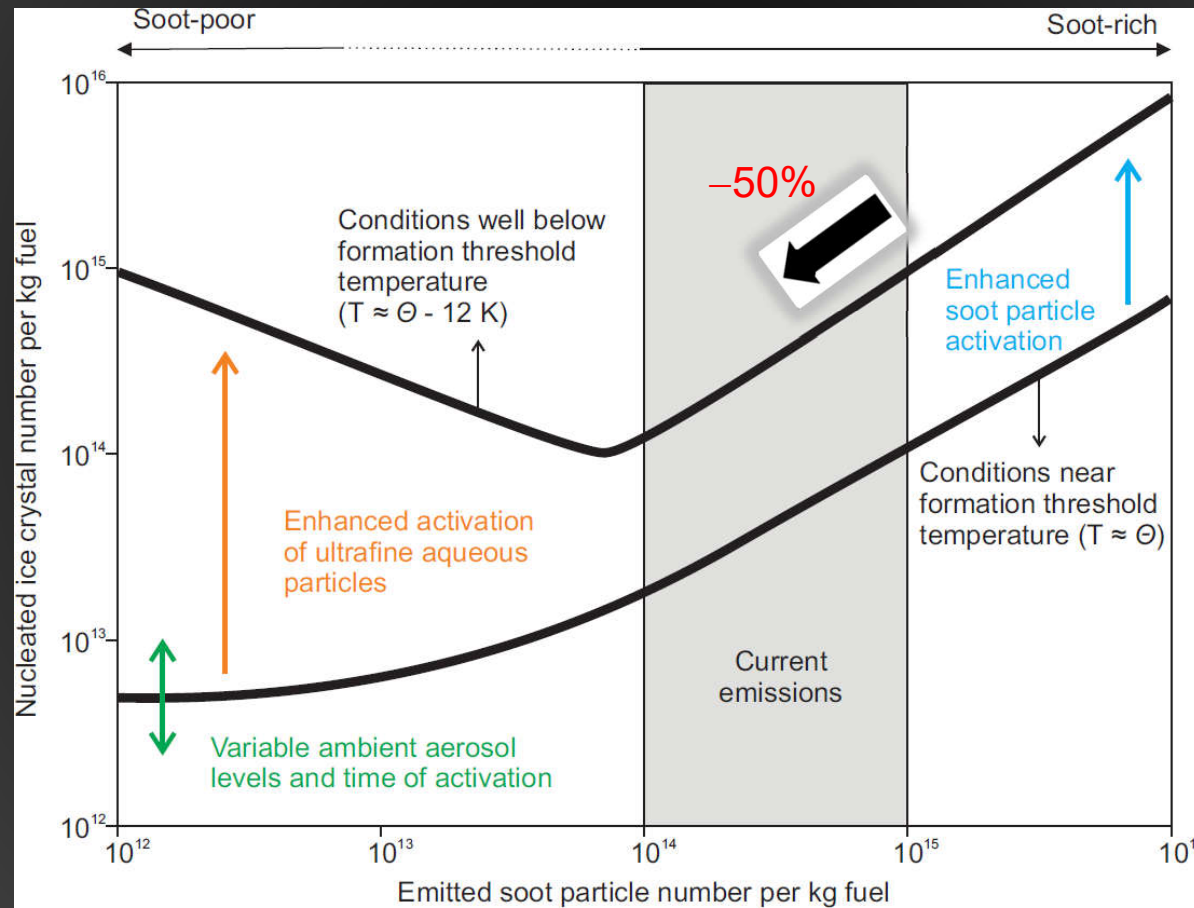
- + reduced sulfur & org./aromatic content (fewer soot & contrail particles)
- + reduced direct AIC RF and potential indirect effect on liquid clouds



- climate model predicts threefold increase in AIC RF by 2050 due to projected increase in fuel use (pre-COVID)
- the same model suggests that the use of fuel blends alone does not decrease aviation's climate impact relative to 2006

- blending mandates (up to 50:50) limit effectiveness
- production to scale (billions of liters p.a.)

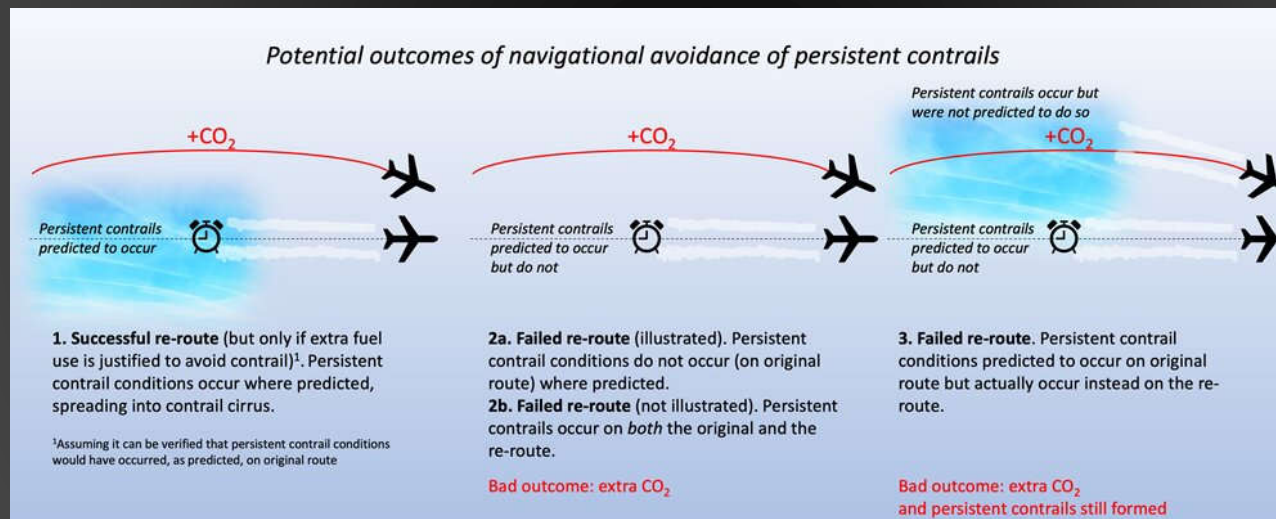
Contrail ice crystal nucleation



Navigational contrail avoidance

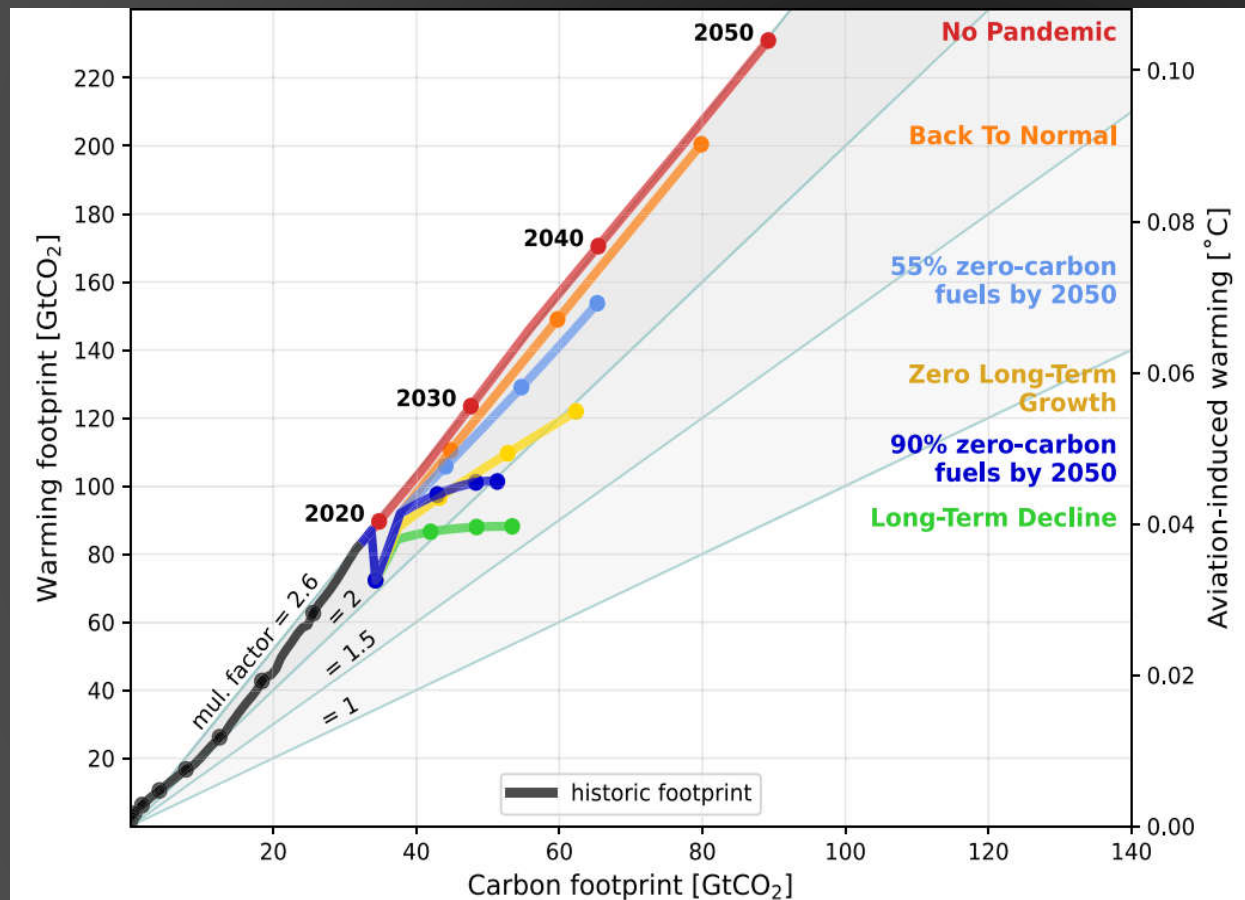
Aircraft re-routing through flight diversions

- + immediate application once operational (ATM) and science issues are solved
- + need to target only subset of flights in AIC outbreak regions



- poor predictability of upper tropospheric ice supersaturation (time, location, extent, magnitude) on the scale of individual contrails
- does not address CO₂ emissions with risk of bad outcomes
- raises governance issues and value judgements

Warming footprint of aviation



Back to normal after COVID19:
aviation caused by 2050 $\approx 0.09^\circ\text{C}$
= up to 1/6 of remaining warming
(COP21 target)

90% zero C fuel use freezes aviation-
induced warming by about 2050;
allowing air traffic to grow, but the
technology is not available to scale

moderate long-term decline in fuel
consumption limits aviation-induced
global warming to 0.04°C (w/o AIC
mitigation)

Take home

- Aviation contributes about 4% to present global warming
 - major drivers are long-lived CO₂ emissions and short-lived AIC forcing
 - Aviation-induced warming can be frozen in by 2050
 - cause less CO₂ emissions (reduces in part also some non-CO₂ effects)
 - business as usual and commit to 90% zero C fuel (reduces CO₂ & AIC)
 - Key scientific research needs related to mitigation of AIC
 - fuel composition and contrail ice crystal numbers
 - upper tropospheric ice-supersaturation
 - global surface temperature response due to AIC forcing
-

References

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